Symbolic decomposition: notable results and comparison evidence

SQUANDER symbolic decomposition project

7 November 2026

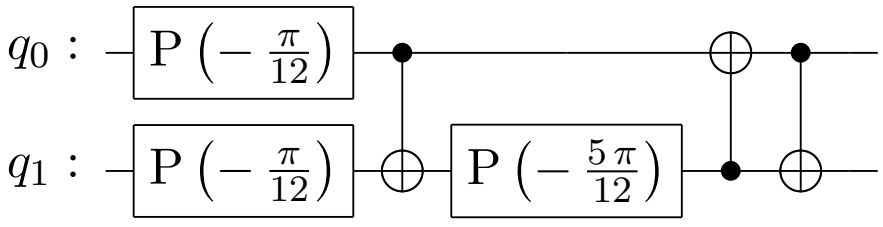
Evidence ledger for a future appendix, not a completed optimality or novelty claim. All counts preserve the recorded native basis, use no ancillas, and fix input/output wires. The objective is entanglers first, the parameter-sensitive score with nonnegative denominator penalties second, and local gate count third. A smaller local count alone does not establish a lower score. New native local constructions include C3SX/iSWAP at 40 locals versus SDK 56, C4X/iSWAP at 78 versus 113, C4X/IBM-CZ at 130 versus 143, and C4X/CZ/Rz/Rx at 69 versus 72. Peres/Rxx/Rz/Ry matches 35. All preserve the native entangler count; these are proved constructions, not claims that the local counts are globally minimum. The version-2 clamp prevents negative-cost identity cycles. An exact audit of all 1,390 retained parameter expressions before the C4X transfer proved their scores unchanged; no existing circuit rerun was needed.

1 Sycamore: a concrete local-gate reduction

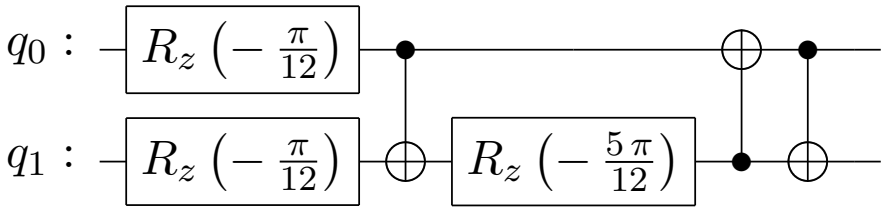
We use SYC = fSim$(\pi/2, \pi/6)$ with odd-parity swap amplitudes $-i$ and $|11\rangle$ phase $e^{-i\pi/6}$. Figure 3 of AbuGhanem and Eleuch [1] contains 15 Rz gates and 12 SX gates: 27 local operations and three CNOTs. This is a manual count of the displayed gate boxes; the two measurements are excluded. The figure has rounded angle labels, so its parameter-sensitive symbolic score and literal scalar phase are not reconstructed or asserted here. Our circuit has three locals and three CNOTs: an 8/9 (88.9%) local-count reduction against that displayed implementation, in the same Rz/SX/CX alphabet projectively. This does not assert that the displayed circuit is the best published local construction.

Construction	Alphabet	E	L	Phase contract
Published Figure 3	Rz/SX/CX	3	27	projective
Our circuit	Rz/SX/CX	3	3	projective
Our circuit	P/H/CX	3	3	literal phase
Qiskit matrix input	cx_u	3	8	scalar metadata
Qiskit matrix input	cx_rz_ry	3	22	scalar metadata
Qiskit matrix input	ibm_cx	3	13	scalar metadata

Literal-phase CNOT/P construction



Same three-angle construction using Rz; scalar shown below



Matched score comparison. In the same Rz/SX/CX basis, the independently synthesized SDK circuit has an exact rational-π lift whose complete identity is proved. The retained score is $\frac{3\log(3)}{5\log(2)} + \frac{9603}{500} \approx 20.157$ for ours and $\frac{3\log(3)}{5\log(2)} + \frac{19809}{500} \approx 40.569$ for the SDK circuit. Both have three CNOTs; the score includes the same entangler contribution and no free physical GP.

The P circuit is exact without a GP operation. Replacing each P by Rz gives SYC $= e^{-7i\pi/24} C_{Rz}$; this scalar is projective metadata. Literal equality in the Rz/SX/CX alphabet is obstructed by its determinant group. The phase polynomial has coefficients $(-\pi/12, -\pi/12, -5\pi/12)$ on $x_0, x_1, x_0 \oplus x_1$. All are nonzero modulo π. Adding 2π to truth-table phases changes these coefficients only by integer multiples of π. Therefore at least three phase gates are necessary within the CNOT/P subalphabet, even if extra CNOTs are allowed. Our three-phase circuit attains that bound. This is not a minimum-local proof over arbitrary U(2) gates or SX-containing circuits.

The independent SDK rows use matrix input, Qiskit 2.2.3, level 3, seed 219, and approximation degree one, followed by native one-qubit block cleanup and phase-sensitive replay. Their circuits and numerical replay errors are stored in the evidence JSON. They are separate from the main baseline, which also lets Qiskit optimize our supplied exact source. That source-fed baseline reproduces the three-local circuit, so its tie is not independent discovery by Qiskit. Where the SDK angles have an exact rational-π lift, the full identity is proved before assigning a symbolic score. Other floating angles are never rationalized and assigned an invented exact score.

2 The eleven entangler improvements

The eleven earlier SDK improvements all concern controlled fermionic SWAP (fFredkin) in different bases. They are completed constructions, separate from the B/SQiSW universal-bound gaps. They are not eleven different gate discoveries. Controlled fermionic SWAP is distinct from ordinary Fredkin; no comparison to the latter's lower bounds is made. Existence of this operation in the literature does not settle optimal decomposition cost [2].

Target	Basis	E ours/SDK	L ours/SDK
fFredkin	cx_rz_ry	5/7	44/39
fFredkin	cx_u	5/7	22/17
fFredkin	cz_rz_rx	5/7	42/19
fFredkin	cz_u	5/7	28/16
fFredkin	ibm_cx	5/7	37/33
fFredkin	ibm_cz	5/7	54/29
fFredkin	ibm_ecr	5/7	53/35
fFredkin	iswap_u	6/8	26/16
fFredkin	xx_rz_ry	5/7	77/35
fFredkin	zx_u	5/7	32/16
fFredkin	zz_u	5/7	37/15

3 Local-gate audit beyond Sycamore

Across 2673 eligible fixed-target/basis comparisons, there are 184 lower local counts, 2477 ties and 1 higher local counts at equal E. Aliases and related basis presets are counted separately. Native one-qubit SDK cleanup removes identity rotations introduced by the manual U-to-ZYZ adapter before counting; savings from that adapter artifact are not credited. These are SDK comparisons, not novelty claims or an exhaustive literature review. The full CSV includes every compared case, including regressions; a separate local-gap CSV is the work list. The table shows the largest local-count savings. E and L are raw physical counts; the exact score is recorded separately in the CSV/JSON. Some larger local counts are deliberate score tradeoffs: the declared objective prefers a lower parameter-sensitive score before counting local gates. Where an exact shorter alternative is present in the archive's history, the CSV records its score and whether the retained circuit has a strictly lower score.

Target	Basis	E ours/SDK	L ours/SDK
C4X	iswap_u	54/54	78/113
C3SX	iswap_u	26/26	40/56
C3X	iswap_u	26/26	42/56
C4X	ibm_cz	36/36	130/143
XX90	xx_rz_ry	1/1	0/12
Peres	ibm_ecr	5/5	31/42
ACX	ibm_cx	1/1	2/12
CT	cx_rz_ry	2/2	3/13
CTdg	cx_rz_ry	2/2	3/13
XX90	ibm_ecr	1/1	6/16
C3X	cx_rz_ry	14/14	19/28
RC3X	iswap_u	12/12	19/28
CNOT	ibm_ecr	1/1	4/12
CSXdg	cx_rz_ry	2/2	6/14
DB	cx_rz_ry	2/2	8/16
SiSWAP	cx_rz_ry	2/2	8/16
XX90	ibm_cx	1/1	7/15
CCX	iswap_u	10/10	16/23
CCY	iswap_u	10/10	16/23
CCZ	iswap_u	10/10	16/23
CSWAP	iswap_u	12/12	20/27
Peres	iswap_u	10/10	16/23
Toffoli	ibm_ecr	6/6	30/37
Toffoli	iswap_u	10/10	16/23
ACX	cx_rz_ry	1/1	4/10

4 Other useful reference results

The separate Clifford atlas contains all 192 phase-resolved one-qubit Clifford targets for each supported local alphabet. H/S has a global shortest-path certificate. H/S/T/Tdg now has an exhaustive exact-word proof covering non-Clifford intermediates: a radius-seven ball of 628 exact matrices certifies all 192 targets, with maximum shortest length 14. Continuous alphabets retain their narrower recorded word-search scope. The exact phase examples include a Cartan kernel with $a = b = c = \pi/8$: three CNOTs projectively, four with literal phase and SU(2)-only Rz/Ry locals.

Likewise, Rxx(π) requires zero entanglers projectively and one with literal phase in the Rxx/Rz/Ry basis. These are certified resource-policy distinctions, not claims of priority over the literature.

5 Certified native-basis constructions

The table records physical entanglers/local gates (E/L). Controlled-phase, CU and native MS rows retain all their real parameters. The controlled-block OSR-two certificate excludes one B or SQiSW pulse whenever the relative block is nonscalar. The square-root SWAP circuits use two B gates and five locals projectively, six with literal phase. The B target uses two SQiSW pulses and three U3 rotations with literal phase. The SQiSW Sycamore motif uses three pulses and a proved parity-block chart. The CZS rows use a two-pulse number-conserving matchgate chart with all three parameters free. These constructions do not assert a global local-gate or parameter-score minimum, or priority over the literature.

Target	Basis	Literal E/L	Projective E/L
CP	sqrt_iswap_u	2/6	2/5
CU	sqrt_iswap_u	2/11	2/10
SYC	sqrt_iswap_u	3/9	3/8
SYC	google_sqrt_iswap	3/9	3/8
SYC	berkeley_b	2/7	2/6
SSWAP	sqrt_iswap_u	3/8	3/7
SSWAP	google_sqrt_iswap	3/8	3/7
SSWAP	berkeley_b	2/6	2/5
DCX	sqrt_iswap_u	2/4	2/4
DCX	google_sqrt_iswap	2/4	2/4
SSWAPdg	sqrt_iswap_u	3/8	3/7
SSWAPdg	google_sqrt_iswap	3/8	3/7
SSWAPdg	berkeley_b	2/6	2/5
B	sqrt_iswap_u	2/3	2/3
B	google_sqrt_iswap	2/4	2/3
CZS	sqrt_iswap_u	2/6	2/6
CZS	google_sqrt_iswap	2/6	2/6
CZS	berkeley_b	2/6	2/6
IonQMS	sqrt_iswap_u	2/16	2/15

The SDK regression count is not the full local optimization work list. The separate native local-layer CSV records 685 phase modes above a local-block count cap, including parameterized families and bases without SDK support. That cap alone does not certify a smaller parameter-sensitive score.

6 Outstanding optimization and comparison work

Classification covers all 327×32 cells in both phase modes. This means a verified construction or a scoped obstruction, not global optimality. The two-B [3] and three-SQiSW [4] bounds identify 10 literal-phase and 10 projective over-cap constructions in the current archive. These are distinct from the local-count gaps above. Further minimum proofs and score optimization are needed even for many circuits below those caps. No competitor-wide dominance or exhaustive novelty claim is justified. UniversalQCompiler has zero executed comparisons pending a suitable Wolfram kernel. Wolfram's free developer engine is not automatically a free paper-benchmark license: its FAQ distinguishes development/testing from producing papers for organizational use. An existing university/site license or written research permission is a possible route [5]; no new access has been obtained.

7 Reproduction and evidence scope

The companion JSON identifies the exact archive and SDK baseline SHA256 hashes, stores the independent Sycamore SDK circuits, and records the published figure's gate-type transcription and image hash. `local_comparisons.csv` contains all eligible comparisons; `local_optimization_gaps.csv` retains regressions. `native_local_layer_opportunities.csv` separately lists native U3/PhasedXZ circuits exceeding one local per wire per pulse layer, plus a literal-phase closure when needed. These are count opportunities, not certified improvements under the parameter-sensitive score. Regenerate with `symbolic_notable_results.py`; compile this source with Tectonic. No numerical SDK sample certifies a free-parameter identity or global minimum.

References

[1] M. AbuGhanem and H. Eleuch, EPJ Quantum Technology 11, 36 (2024), doi:10.1140/epjqt/s40507-024-00248-8, Figure 3.

[2] npj Quantum Information (2023), doi:10.1038/s41534-023-00711-x.

[3] J. Zhang et al., Phys. Rev. Lett. 93, 020502 (2004), arXiv:quant-ph/0312193.

[4] C. Huang et al., Quantum Instruction Set Design for Performance, arXiv:2105.06074.

[5] Wolfram Engine FAQ, licensing and terms of use, `https://www.wolfram.com/engine/faq/`, checked 6 November 2026.